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Effects of Acetylshikonin on the Infection and Replication of Coxsackievirus A16 *in Vitro* and *in Vivo*

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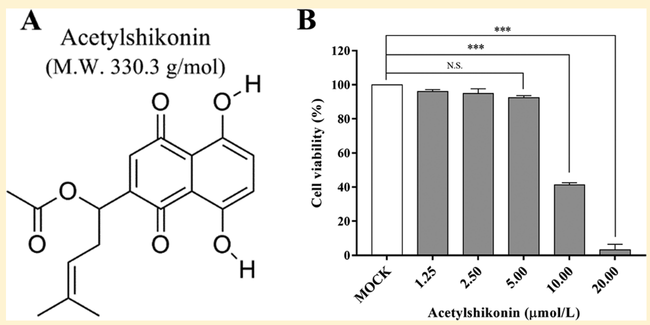
Supporting Information

ABSTRACT: Coxsackievirus A16 (CVA16) is one of the most prevalent enteroviral pathogens associated with hand, foot, and mouth disease. In the present study, we have investigated (1) whether the bioactive compound acetylshikonin (AS) inhibits CVA16 infection *in vitro* and *in vivo* and (2) the potential antiviral mechanism(s). The results suggest that AS is nontoxic at concentrations of up to 5 μmol/L and could directly inactivate virus particles at relatively low concentrations (0.08 μmol/L), thereby rendering CVA16 incapable of cellular entry. Correspondingly, the expression of viral RNA *in vitro* was also reduced 100-fold ($P < 0.05$) when compared to infected, untreated controls. Results from a CVA16-infected neonatal mouse model indicate that, in comparison to the virus-infected, untreated group, body weights of the mice in the virus-infected, compound-treated group increased more steadily with less severe clinical symptoms. In addition, viral loads in internal organs significantly decreased in treated animals, concomitantly with both reduced pathology and diminished expression of the proinflammatory cytokines IFN-γ and IL-6. In conclusion, AS exerted an inhibitory effect on CVA16 infection *in vitro* and *in vivo*. Our study provides a basis for further investigations of AS-type compounds to develop therapeutics to mitigate CVA-associated disease in children.



Coxsackievirus (CVA16) is a nonenveloped, positive-polarity, single-stranded RNA virus member of human enterovirus species A (HEV-A) within the genus *Enterovirus*, family *Picornaviridae*.[1] The CVA16 linear viral genome of approximately 7.4 kb possesses a single open reading frame (ORF), which encodes a large polyprotein that is cleaved by viral and host-derived proteases into three protein precursors: P1, P2, and P3. P1 is subsequently digested enzymatically into four structural proteins, VP1, VP2, VP3, and VP4, which are the main components of the viral capsid. VP1 is the primary antigenic determinant, and P2 and P3 encode nonstructural proteins.[2,3] In the past decade, the prevalence of CVA16-associated hand, foot, and mouth disease (HFMD) outbreaks in children has increased in China and other countries,[4−6] and prevalence has even exceeded that of enterovirus A71 (EVA71) in some years.[7,8] Previously, we have studied a large cohort of 839 483 patients (including >370 000 HFMD cases) in Shandong Province between 2009 and 2016 and found that EVA71 and CVA16 were the principal pathogens, with CVA16 predominant in 2010, 2012, and 2016, accounting for 37%, 45%, and 53% of the total number of laboratory-diagnosed HFMD cases, respectively.[9] In addition to HFMD, CVA16 can also cause severe complications such as pulmonary edema, myocarditis, intractable shock,[10] and rhombencephalitis,[11] which are associated with high case fatality rates in children.[12] Nevertheless, no vaccines or antivirals specific for the prevention and treatment of CVA16 have been licensed to date.

At present, the clinical treatment of CVA16-associated HFMD principally involves curtailing uncontrolled inflammatory responses and symptomatic treatments with glucocorticoids and ribavirin, rather than employing EV-specific antivirals that inhibit infection/replication. However, these compounds have serious side effects; glucocorticoids can impair innate immunity by inhibiting the activity of multiple immune effectors,[13] and the nucleoside analogue ribavirin can result in disorders of the hematopoietic system.[14] It has been

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reported that acetylshikonin (AS), isolated from the Chinese medicine *Zicao*, at purities of 98.4% as detected by high-performance liquid chromatography,[15] possesses potent antiviral,[16,17] antitumor,[18,19] and anti-inflammatory effects.[20,21] For example, in chronic hepatitis B, AS induces apoptosis in HBx-expressing hepatocarcinoma via induction of endoplasmic reticulum (ER) stress to achieve its antiviral effects.[22] In addition, Chen et al. has discovered that AS mainly down-regulated surface expression of CCR5, a primary HIV-1 co-receptor on macrophages, thereby inhibiting retrovirus infection.[23] No prior studies on the pharmacological effects of AS against enteroviruses have been reported to date.

In the present study, the inhibitory effects of AS on CVA16 replication *in vitro* in cultured cells and *in vivo* using a neonatal mouse infection model were evaluated. Quantification of viral loads, pathology, and expression of proinflammatory cytokines were determined, the survival rate of infected neonates was monitored, and the anti-CVA16 mechanism of AS was investigated. We provide evidence that AS is a promising compound candidate to develop as an antiviral for the treatment of CVA16-associated HFMD cases and potentially other picornavirus infections.

■ RESULTS AND DISCUSSION

Determination of the Cytotoxicity of AS. A range of concentrations of AS were added to human rhabdomyosarcoma (RD) cells, and the cell viability was measured at 24 h by the CCK-8 method. We determined that when AS concentration was <5 μmol/L, the cell survival rate was >90% (Figure 1). With increased compound concentration at 10 and 20 μmol/L, cell death was significantly more pronounced, which indicated the compound was more toxic to the cells at these concentrations. Thus, the maximum nontoxic concentration of AS was set at 5 μmol/L.

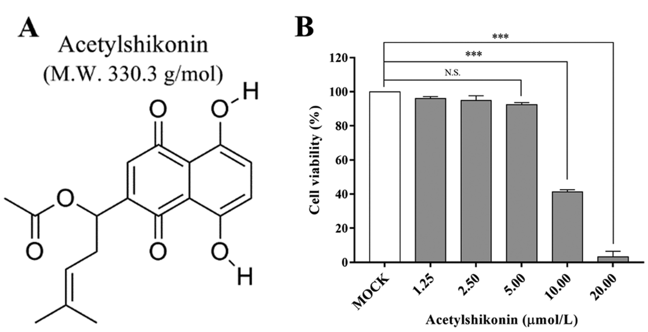

Figure 1. Determination of the cytotoxicity of AS to RD cells. (A) Chemical structure and molecular weight of AS. (B) RD cells were incubated with a range of AS concentrations for 24 h, and then cell viability was determined by the CCK-8 method. The maintenance solution (MEM supplemented with 2% FBS and 1% penicillin–streptomycin) served as mock-infected controls. Data shown are the mean ± SD from three independent experiments. Statistical significance was compared between the test groups and the mock-infected group (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$).

Determination of Anti-CVA16 Activity of AS. AS at a range of different concentrations (from 5 to 0.01 μmol/L) was 1:1 mixed with CVA16 strain TA271 (10^4 $TCID_{50}$/0.1 mL), preincubated in a 5% CO_2 incubator at 37 °C for 2 h, and then inoculated onto RD monolayers. The cell survival rate and virus infection rate were compared at different compound concentrations using the CCK-8 cell viability assay, which allowed the effective anti-CVA16 compound concentrations to be calculated. The 50% cytotoxicity concentration (CC_{50}) and the 50% effective concentration (EC_{50}) of the compound were calculated. As shown in Figure 2A,B and Table 1, AS and ribavirin respectively showed CC_{50} values of 9.4 and 1485 μmol/L. The antiviral effect of the two compounds is presented in Figure 2C and D, and both AS and ribavirin showed a dose-dependent antiviral effect with EC_{50} values of 0.04 and 143.61 μmol/L, respectively. Comparison of the selectivity index (SI) of AS and ribavirin (the positive drug control) in Table 1 clearly showed that the AS SI value (235) was much greater than the ribavirin SI value (10.34), which suggested that AS is likely to be both a safer and more efficacious compound against CVA16. As shown in Figure 2C, AS reduced CVA16-induced cytopathic effect (CPE) with maximum inhibition rates of 80% at the lowest concentration of 0.08 μmol/L, which was therefore the concentration chosen to be used for further research on the antiviral mechanisms of AS.

AS Inhibited CVA16 in the Adsorption/Entry Stage. In order to determine the stage of the viral life cycle when AS inhibits virus infection, the test compound was added at four different stages: preinfection, adsorption/entry, replication, and release. CPE was monitored and cell culture supernatants were collected to determine viral loads by real-time PCR. The morphology of the RD monolayers at 48 hours postinfection (hpi) from the different groups is shown in Figure 3. The cells in the (untreated) control group are shown in Figure 3A for comparison. Strikingly, compared to the viral infection group (Figure 3B), the number of viable cells counted in the adsorption/entry period (Figure 3D) was relatively higher, and the cell monolayer was more comparable morphologically to the cell control. In contrast, when the compounds were added at the stage of viral preinfection (Figure 3C), replication (Figure 3E), and release (Figure 3F), the cells appeared shrunken, rounded, detached, and fragmented, indicative of cytotoxicity. Intracytoplasmic granules were also increased and showed refractive indices. The cell morphology in Figure 3C, E, and F showed no discernible differences from those in the viral infection control group. Notably, viral loads were also consistent with the previous microscopic observations with significantly lower viral genome copies determined when AS was added at the adsorption/entry stage but no significant differences at other stages (preinfection, replication, and release) of the life cycle when compared to the untreated virus-infected RD cells (Figure 3G). These results indicated that AS exerted anti-CVA16 activity in the adsorption/entry stage but was ineffective in the virus preinfection, replication, and release stages.

AS Inactivates Free Viral Particles to Inhibit CVA16 Infections. The previous experiment provided evidence that AS could potentially block CVA16 from adsorbing to the cell surface and/or entry into the cell. We speculated that the compound could directly interact with and inactivate the virus particles, thereby inhibiting CVA16 infection, and a viral inactivation assay was performed as described in the Experimental Section to test our hypothesis. As shown in Figure 4, it was evident that the number of viral genome copies in the noninactivated group was significantly higher at 24 and 48 hpi when compared with the inactivated group with AS–virus preincubated for 2 h ($P < 0.05$) with a reduction of >2 $\log_{10}$ titers at 24 hpi. These findings indicated that AS directly inactivated CVA16 viral particles to inhibit infection. In order

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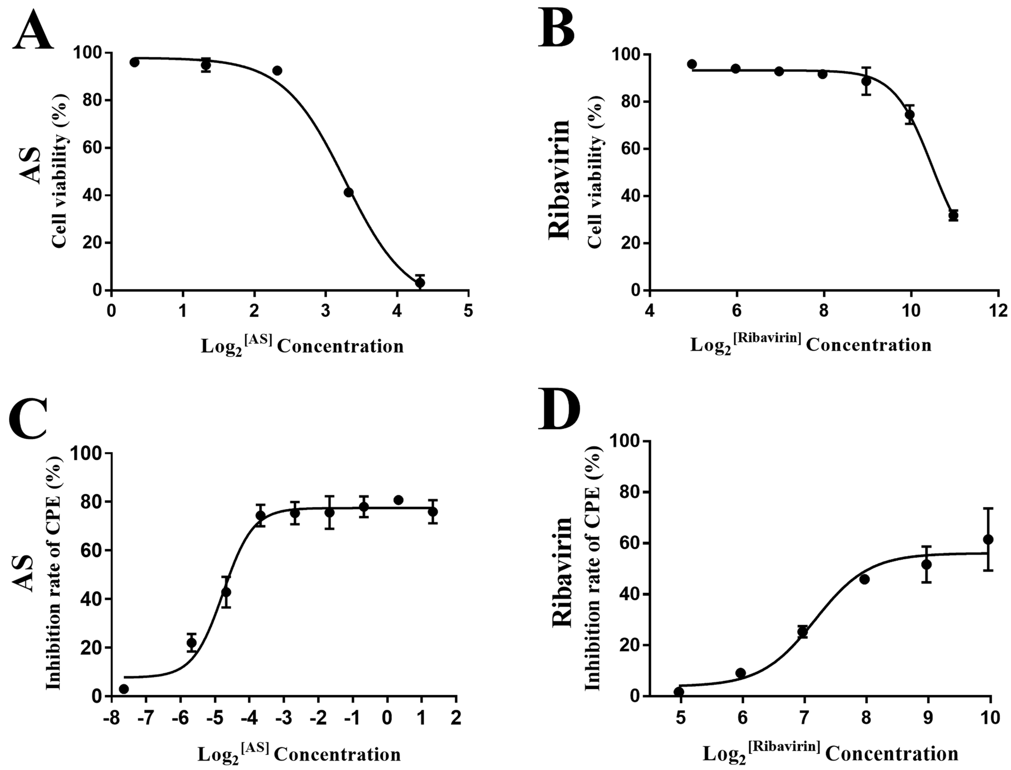

Figure 2. Determination of the activity of AS against CVA16. RD cells were inoculated with the mixture of the test compound at different concentrations and CVA16 (10^4 $TCID_{50}$/0.1 mL) that was preincubated for 2 h at 37 °C, and then the cell viability was analyzed using the CCK-8 cell viability assay. Uninfected media (2% MEM) served as mock-infected controls. Data shown are the mean ± SD from three independent experiments.

Table 1. Cytotoxicity and Antiviral Activities of the Compounds against CVA16 in RD Cells[a]

	cytotoxicity	antiviral activity	
compound	CC_{50} (μmol/L)	EC_{50}[b] (μmol/L)	SI[c]
AS	9.4	0.04	235
ribavirin	1485	143.61	10.34

[a]Results are presented as mean values obtained from three independent experiments. [b]The concentration at which the compound inhibition rate of CPE reaches halfway between the baseline and maximum. [c]The selectivity index (SI) value represents the ratio of CC_{50}/EC_{50} for the compound.

to further substantiate these findings, we selected an additional three heterogeneous CVA16 strains (LW2, LW614, and 16038; Figure S1) that were distantly genetically related to the strain TA271 we employed previously so as to determine whether AS also exhibited a more broad-spectrum antiviral effect on these CVA16 strains. As shown in Figure S2, the CVA16 viral loads of all the strains in the inactivated group were significantly decreased in comparison to the positive control group ($P < 0.05$), suggesting that AS could exert an inhibitory effect on diverse CVA16 strains.

Changes in Body Weight, Clinical Score, and Survival Rate of Neonatal Mice Infected with CVA16 after Different AS Dosing Regimens. Five-day-old ICR suckling mice were treated with AS (treatment group and prevention group) to evaluate the therapeutic and preventative effects of the test compound on CVA16 infection. A negative control group and a positive control group were also established in parallel. The survival rate, body weight, and clinical score of the neonatal mice were monitored daily. As shown in Figure 5, all the neonatal negative control mice grew healthily, and their body weights and clinical signs were normal. In contrast, the positive control group began to die at 4 days postinfection (dpi), and all died 6 days after CVA16 infection. In the treatment and prevention groups the mice began to die later (6 and 7 dpi), and the final survival rates were 50% and 70% (Figure 5A), respectively. In addition, compared to the positive control group, whose body weight dropped sharply at 5 dpi, both the treatment and the prevention groups exhibited a gradual increase in body weight (Figure 5B). The mean clinical score of the positive group rapidly rose at 1 dpi, and as the number of days postinfection increased, single hind limb muscle paralysis increased, followed by bilateral hind limb muscle paralysis and moribund presentation until death of all animals by 6 dpi. The symptoms in the treatment and prevention groups appeared later, and the average clinical score was lower. The final clinical score of the treatment group was <3 and that of the prevention group was <2 (Figure 5C). In summary, AS had a more pronounced anti-CVA16 effect in neonates, and the impact of the compound on the prevention of viral infection was greater than that of the treatment group.

Determination of the Viral Loads in Organs of the Neonates Infected with CVA16 after Different AS Dosing Regimens. After the 5-day-old neonatal mice were challenged with CVA16 strain TA271 with different AS dosing regimens, the viral load in each organ was measured at 1, 3, and 5 dpi. It was apparent that the viral loads varied between different organs, and the viral loads of all organs rapidly increased from 1 to 3 dpi; however, the increase rate slowed significantly after 3 dpi (Figure 6). In the collected organs, the viral load of the hind limb muscles was highest in the positive

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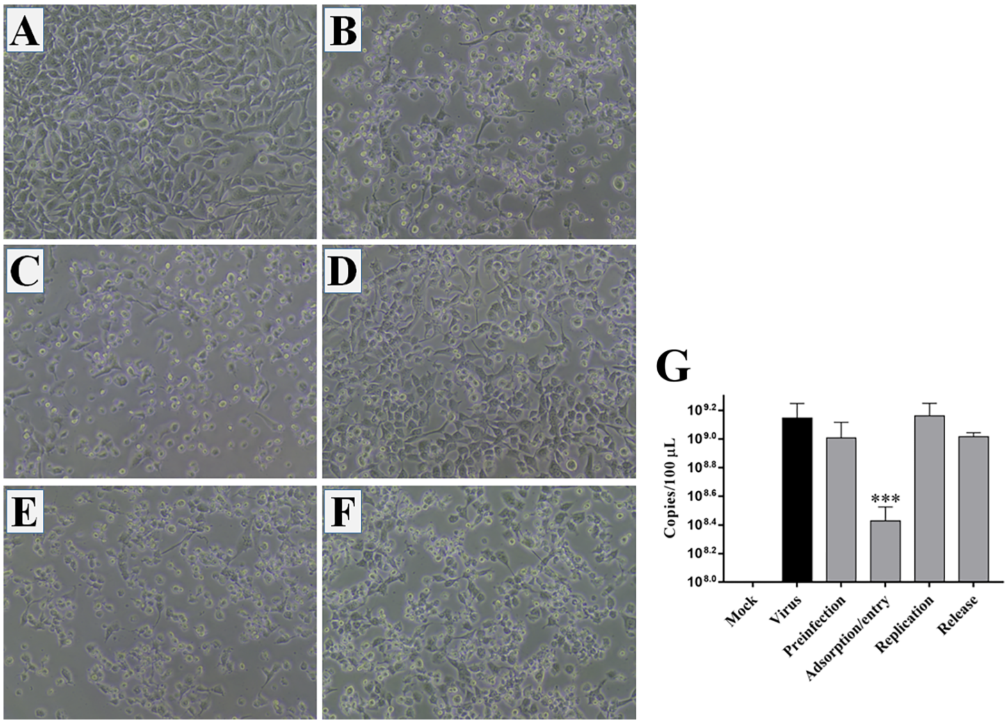

Figure 3. Determination of the role of AS in inhibiting CVA16. (A) Cell control group: uninfected RD cells; (B) virus infective group: CVA16 (10^4 $TCID_{50}$/0.1 mL); (C) virus preinfection period; (D) virus adsorption/entry period; (E) virus replication period; (F) virus release period. Magnification, 100× (A–F). (G) RD cell culture supernatants were collected from the experimental group and the control group at 48 hpi, and, subsequently, total RNA was extracted and the viral load was determined by real-time PCR. Data shown are the mean ± SD from three independent experiments. Statistical significance was compared between the test groups and the virus infective group (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$).

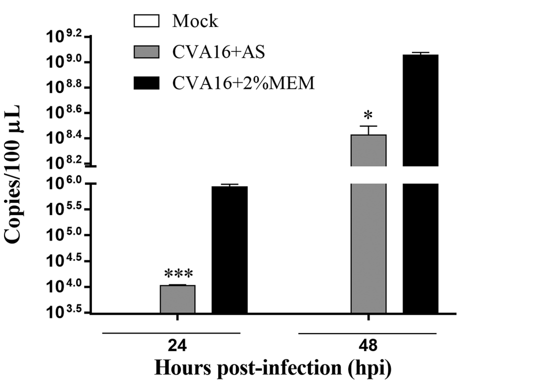

Figure 4. AS directly inactivated CVA16 particles. CVA16 (10^4 $TCID_{50}$/0.1 mL) and AS (0.08 μmol/L) or 2% MEM were co-incubated at 37 °C for 2 h, and the two mixtures were inoculated onto RD cells, separately. Total RNA was extracted from the cell culture supernatant at 24 and 48 h, and viral RNA was measured by real-time PCR. Data shown are the mean ± SD from three independent experiments. Statistical significance was compared between the inactivated group and the noninactivated group (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$). Copies in the mock group were too low, and data are not shown in the figure.

control, untreated group by 5 dpi (>10^7 genome copies/mg), followed by the lungs and spinal muscles (each >10^6 genome copies/mg), with slightly lower viral titer of CVA16 in cardiac tissue (10^6 genome copies/mg). The viral load in each organ or tissue from both the treatment and prevention groups were significantly reduced at 3 and 5 dpi compared with those of the positive control group ($P < 0.05$). Notably, the rate of viral replication was also relatively reduced each day after infection in both the treatment and prevention groups when compared to the (untreated) positive control group. These findings indicated that AS could inhibit the replication of CVA16 in the suckling mice.

Histopathological Analysis of Various Organs of the Neonatal Mice Infected with CVA16 after Different Dosing Regimens. The heart, lungs, hind limbs, and skeletal muscles of the suckling mice in each treatment group and positive control group were analyzed to determine histopathological features. The results showed that all the mice in the challenge groups had pathological changes to differing degrees, manifested as increased myocardial cells and myocardial fiber rupture (Figure 7A), and alveolar epithelia was significantly thickened with infiltrating inflammatory cells apparent (Figure 7B). Both hind limb and spinal muscles were also hemorrhaged and severely necrotic. Muscle fibers were ruptured, muscle fiber cells were lysed, and muscle nuclei were swollen and contracted and mixed with inflammatory cells (Figure 7C and D). In the treatment group (Figure 7E–H), the appearance of lesions in each tissue was milder, with relatively less inflammatory cell infiltration, less breakage of muscle fibers, milder alveolar fibrosis, and milder alveolar wall thickening. In the prevention group (Figure 7I–M), the histopathological lesions in the mice were not significant; however, there was more inflammatory cell proliferation in the spinal muscles than the other three organs examined in the treatment group. Taken together, the findings suggest that AS is capable of inhibiting CVA16 replication and reducing histopathological changes when administered as either a therapeutic or a preventative compound.

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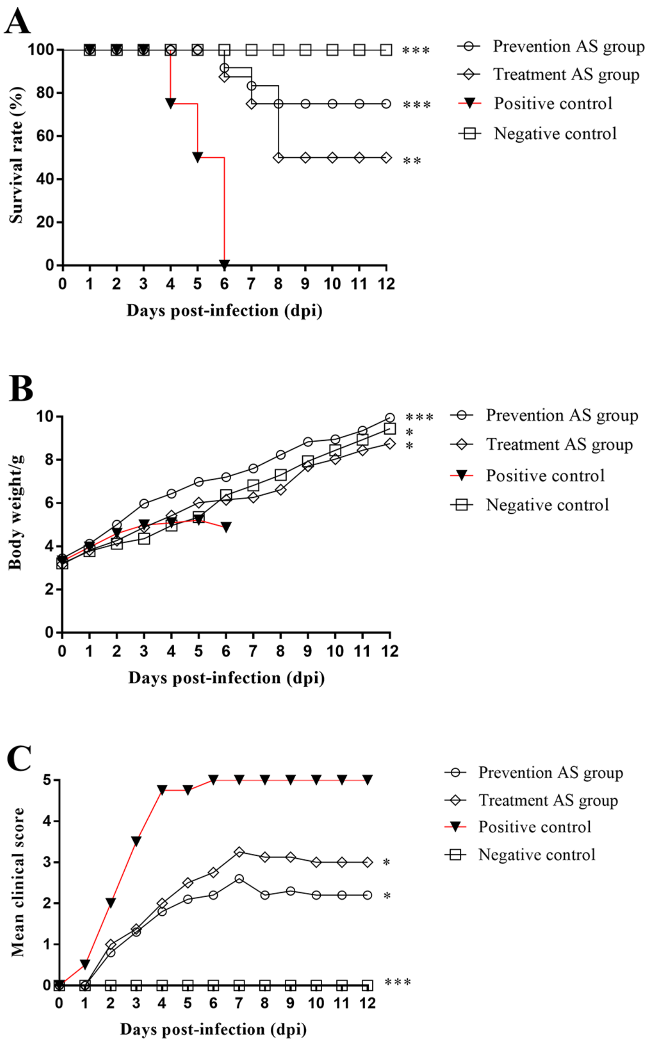

Figure 5. Dynamic changes of body weight, clinical scores, and survival rates in CVA16-infected mice by different AS dosing regimens. (A) Changes in survival rates of the CVA16-infected neonatal mice infected with different AS dosing regimens. (B) Changes in body weight of the CVA16-infected neonatal mice with different AS dosing regimens. (C) Changes in clinical scores of the CVA16-infected neonatal mice with different AS dosing regimens. Statistical significance was compared between the test groups and the positive control (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$).

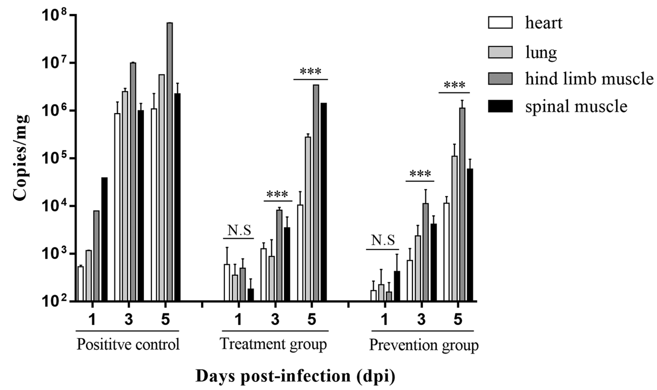

Figure 6. Monitoring the viral load in various organs of the neonatal mice infected with CVA16 after different AS dosing regimens. Five-day-old ICR mice were i.m. inoculated with CVA16 through various AS dosing regimens. The viral loads in the heart, lung, hind limb muscle, and spinal muscle from the infected mice were measured by real-time PCR. Samples were collected at the indicated time points. The results represent the mean virus load (copies) per milligram of tissue. Data shown are the mean ± SD from three independent experiments. Statistical significance was compared for the survival rate (12 dpi), the weight (6 dpi), and the mean clinical score (12 dpi) between the test groups and the positive control (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$).

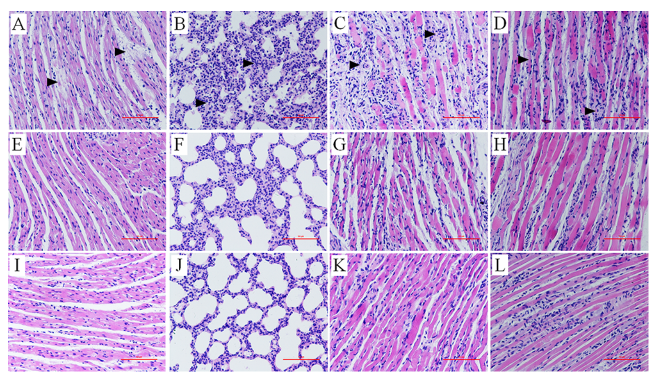

Figure 7. HE staining of various organs of the CVA16-infected neonatal mice after different AS dosing regimens. (A−D) HE staining of heart, lung, hind limb muscle, and spinal muscle at 5 dpi in the positive control group; (E−H) HE staining of heart, lung, hind limb muscle, and spinal muscle at 5 dpi in the treatment group; (I−M) HE staining of heart, lung, hind limb muscle, and spinal muscle at 5 dpi in the prevention group. Magnification, 100×, scale bar 100 μm (B, F, G); 200×, scale bar 200 μm (others). All experiments were repeated three times.

Detection of Plasma Proinflammatory Cytokines in Neonatal Mice Infected with CVA16 after Different AS Dosing Regimens. Five-day-old mice in either the treatment or the prevention groups had blood taken from the heart at different time points (3 and 5 dpi); plasma was isolated and the expression levels of IFN-γ and IL-6 were detected by ELISA. As shown in Figure 8, compared with the negative control group, the level of IFN-γ in the infection group markedly increased 3 dpi. Interestingly, the expression level of IFN-γ in the treatment and prevention groups was dramatically lower than that in the positive control group ($P < 0.05$). Although the levels of proinflammatory cytokines increased 5 dpi in the treated groups, it was still lower than that in the positive control group. Therefore, AS inhibited the high expression of IFN-γ at 3 dpi when compared to untreated controls. Although the expression level of IL-6 in the positive group, the treatment group, and the prevention group was not significantly different at 3 dpi, the IL-6 level in the infected group was as high as 440 pg/mL, and the IL-6 expression levels in the treatment and the prevention groups were significantly reduced at 5 dpi ($P < 0.05$). Besides, by comparing the expression levels of IL-6 and IFN-γ in the compound control (AS treated) and that of negative control mice, it was found that there was no significant difference in IL-6 and IFN-γ levels between the two groups on the same day, indicating that AS could not directly affect the expression levels of IL-6 and IFN-γ before the suckling mice were infected with CVA16. In brief, the results demonstrated that AS could indirectly reduce the IL-6 level and inhibit the expression of uncontrolled proinflammatory reactions following CVA16 infection.

The outbreak of HFMD in children is associated with a number of enterovirus serotypes, with the two most prevalent viruses, identified from fecal and cerebrospinal fluid samples of

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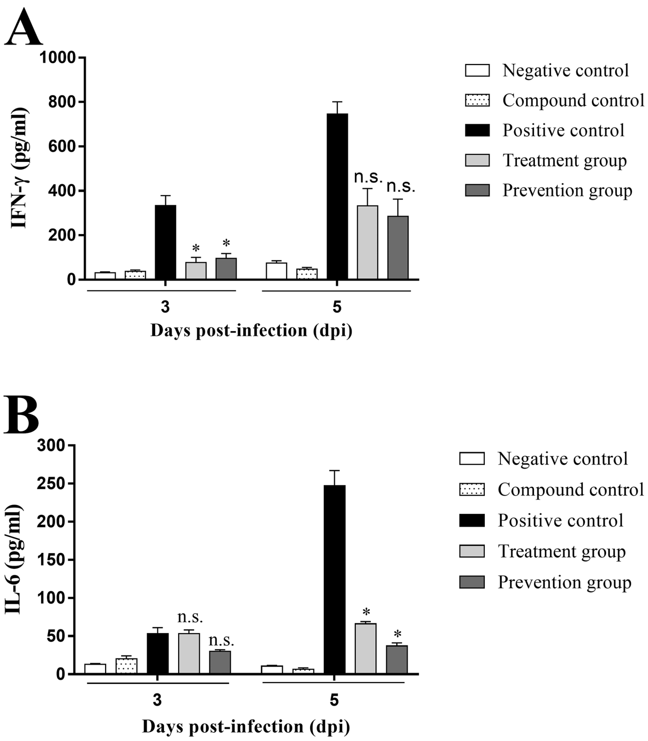

Figure 8. Determination of plasma proinflammatory cytokines in CVA16-infected neonatal mice after different AS dosing regimens. (A) Changes in plasma IFN-γ expression 3 and 5 dpi after CVA16 infection in 5-day-old pups of suckling mice. (B) Changes of IL-6 expression 3 and 5 dpi after CVA16 infection in the 5-day-old pups. Data shown are the mean ± SD from three independent experiments. Statistical significance was compared between the test groups and the positive control (*$P \leq 0.05$, **$P \leq 0.01$, ***$P \leq 0.001$).

HFMD cases, being CVA16 and EVA71.[24] It has been reported previously that humans can also be co-infected with both CVA16/EVA71,[25] and the co-infection phenomenon can increase the possibility of intertypic recombination events.[26] Notably, co-circulation of CVA16 and EVA71 has increased the possibility of co-infections and genetic recombination, which further increases the difficulty associated with controlling HFMD epidemics without licensed antiviral treatment options.[27] At present, the EVA71-inactivated vaccine has been licensed for use;[28,29] however, there is currently no licensed vaccine that can effectively prevent CVA16 infection.[30] It is therefore imperative to develop effective antiviral compounds against CVA16 (and other HFMD-associated pathogens) to enable both the treatment and prevention of infection. Although broad-spectrum antiviral compounds, such as glucocorticoids and ribavirin, are currently in use, they often cause serious side effects to immunity and hematopoiesis in humans.[13,14] Yu and co-workers systematically evaluated the clinical efficacy and safety for HFMD treatment by *Yinqiao decoction*, a traditional Chinese medicine formulation, and presented evidence for its therapeutic efficacy.[31]

As mentioned earlier, AS has some antiviral properties; however, its precise mechanism of action(s) remains unclear. In the present study, *in vitro* cell experiments provided evidence that AS could significantly inactivate CVA16 virus particles, thereby rendering them incapable of infecting target cells. Ma and colleagues have demonstrated that propionyl shikonin binds to tobacco mosaic virus, inactivating the virion.[32] Shikonin has also been demonstrated to effectively inhibit HIV endocytosis by down-regulating the expression of HIV-1 co-receptor chemokines and blocking the binding of HIV-1 to its receptor.[23] Interestingly, a propanoate ester of shikonin (PMM-034) is able to reduce the replication efficiency of influenza A (H1N1) in a dose-dependent manner in Madin-Darby canine kidney (MDCK) cells and human lung carcinoma (A549) cells[33] and can also significantly inhibit the proliferation of dsDNA viruses, such as adenovirus type 3 (AdV3), in host cells.[34] Thus, shikonin and its derivatives have a broad spectrum of antiviral activities that can be achieved by different antiviral pathways or mechanisms as described previously. Our study provides evidence that AS directly inactivated CVA16 viral particles and effectively prevented the virus from adsorbing or entering the cell. We speculate that AS competitively binds to the hydrophobic pockets of VP1 on the virus surface, thereby preventing the interaction between CVA16 and one of cell surface attachment receptors, such as the heparan sulfate glycosaminoglycans.[35]

It has been reported that CVA16 can cause significant myocardial necrosis and breakage of myofilament fibers in a suckling mouse infection model.[36] In the present study, we found that AS could provide a certain degree of protection for suckling mice challenged with a lethal dose of CVA16. Interestingly, we studied the protective and therapeutic effects of three doses of compounds including large (10 mg/kg), medium (2 mg/kg), and small (0.4 mg/kg) on CVA16-infected suckling mice. As presented in Figure S3, AS had a survival rate of 80% by preventive action at 10 mg/kg, and the therapeutic survival rate was correspondingly low (50%), which indicated that AS has a better antiviral effect as a preventive than a therapeutic drug. Moreover, AS dramatically reduced the replication of the virus in the myocardium, skeletal muscle, and other organ tissues, thus alleviating uncontrolled inflammatory reactions and reducing histopathological changes. Shikonin has been shown to significantly improve the dysfunction of cerebral ischemic nerve injury, hydrocephalus, and the integrity of the blood−brain barrier (BBB) by modulating inflammatory responses and promoting the permeability of the BBB, thus protecting the brain from ischemic injury.[37] It was also reported that shikonin could reduce the release of primary macrophage cells in mice and block the translocation of NF-κB-P65 from the cytoplasm to the nucleus to lessen proteasomal-related activities. Under these circumstances, it has been suggested that shikonin acts as a proteasome inhibitor under inflammatory conditions[38] and further suppresses the secondary inflammatory reaction to protect cardiomyocytes. Viral infection induces high expression of a variety of proinflammatory cytokines ("cytokine storms"), causing damage to multiple tissues and organs, thereby leading to severe disease.[39] Prior studies have demonstrated that IL-6 is highly expressed in patients with HFMD, and as the concentration of IL-6 rises, disease severity also increases.[40] Hence, it is vital to identify compounds that can down-regulate proinflammatory factors to diminish the pathology associated with HFMD disease, particularly in pediatric cases with severe neurological and cardiopulmonary sequelae. Coincidentally, Dai and co-workers have provided evidence that shikonin exerts anti-inflammatory effects by inhibiting the expression of Th1 cytokines (IL-6 predominant) and then suppressed the production of IL-6.[41] Evidence exists that shikonin can inhibit both the secretion and expression of IL-6 and IL-8 by significantly enhancing the expression of lncRNA-

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NR024118.[42] These previous findings are consistent with the data presented in the present study where concentrations of IL-6 and IFN-γ were significantly lower in the AS-treated, CVA16-infected groups than those in the untreated, virus-infected control group. This occurred concomitantly with a decrease in the incidence of severe pathology, indicating that a likely antiviral mechanism of action of AS is to reduce disease severity by diminishing uncontrolled proinflammatory responses.

In conclusion, this study has provided evidence that AS could effectively protect cells from CVA16 by directly inactivating virus particles and blocking the process of viral absorption/entry *in vitro* and, furthermore, that the compound could reduce CVA16 infection in a neonatal murine model, relieve symptoms, and alleviate disease-associated proinflammatory responses. Due to the large number and diversity of enterovirus species, whether AS has the same effects on other serotypes needs to be further investigated. Our study lays a foundation for the development of AS, and derivatives, as antiviral therapies for future clinical application for the treatment of HFMD-associated pathogens to limit pediatric disease.

■ EXPERIMENTAL SECTION

Test Compounds. Acetylshikonin (purity ≥98.4%; molecular weight 330.3 g/mol) was purchased from Sigma, Beijing, China, and dissolved in dimethyl sulfoxide (DMSO, Beijing, China) to a stock concentration of 10 mmol/L and stored at −20 °C until further use.

Cells and Viruses. Human rhabdomyosarcoma cells, supplied by the Shandong Center for Disease Control and Prevention, were cultured in minimum essential medium (MEM, Gibco, Suzhou, China) supplemented with 10% fetal bovine serum (FBS, Gibco, Suzhou, China) and 1% penicillin−streptomycin mixture (Solarbio, Beijing, China). The CVA16 strain TA271 (TA271/Shandong/China/2015) employed in the present study was isolated in 2015 from a stool sample from a pediatric HFMD case. After isolation and expansion three times in RD cells *in vitro*, the virus titer was determined to be 10^7 $TCID_{50}$/0.1 mL and stored at −80 °C until further use. The whole genome was amplified by RT-PCR and sequenced, as described previously.[43] The CVA16 strain TA271 genome sequence was deposited in NCBI (GenBank No. MG674827), and phylogenetic analysis confirmed that the genotype of this strain is B1b.

Cytotoxicity Assay. Near 90% confluent RD cells were stained with 0.04% trypan blue dye, seeded in 96-well plates (5000 cells/well), and incubated for 24 h at 37 °C under 5% CO_2. AS was 2-fold serially diluted from the 10 μmol/L stock solution with maintenance solution (MEM plus 2% FBS and 1% penicillin−streptomycin). RD cells were treated with the serially diluted range of compound concentrations and incubated for 12 h at 37 °C with 5% CO_2. Five replicates were prepared for each concentration. A cell control (no compounds added) and a blank control were also established. To each well was added 10 μL of CCK-8 Reagent (Multi Sciences, Hangzhou, China). After the cell plates were incubated at 37 °C for 1.5 h, the absorbance (A) at 450 nm was recorded using a microplate reader (ELX-800, BIOTEK, China), and the cell survival rate was calculated according to the following equation: Cell survival rate (%) = $(A_{\text{experimental group}} - A_{\text{blank group}}/A_{\text{cell group}} - A_{\text{blank group}}) \times 100\%$.

Antiviral Assay. A 2-fold serially diluted range of AS concentrations from 5 μmol/L (the maximum nontoxic concentration) to 0.01 μmol/L were mixed with an equal volume of CVA16 (10^4 $TCID_{50}$/0.1 mL) and preincubated at 37 °C for 2 h prior to adsorption. The compound−virus mixture was then inoculated onto 80% confluent RD monolayers with three replicates at each compound concentration. As a positive drug control, RD cells were also treated with ribavirin in 2% MEM at serially diluted concentrations 1 hpi. The treatment groups above together with uninfected (cell control) and untreated (virus control) groups were incubated for 24 h at 37 °C under 5% CO_2. The absorbance of the cells in different treatment groups was measured according to the manufacturer's instructions of the CCK-8 cell viability kit (Multi Sciences, Hangzhou, China). The cell survival rate and the virus inhibition rate were calculated to identify the optimal compound concentration (0.08 μmol/L) for the antiviral effect. Also, the EC_{50}, referring to the concentration of a drug that induces a response halfway between the baseline and maximal after a specified exposure time, was calculated for the compounds by regression analysis using GraphPad Prism 7.0, as reported previously.[44]

Time-of-Compound-Function Assays. *Virus Preinfection Stage.* RD cells cultured in 24-well plates were treated with AS (0.08 μmol/L) and incubated at 37 °C for 3 h, and then cells were washed with phosphate-buffered saline (PBS) and infected with CVA16 (10^4 $TCID_{50}$/0.1 mL) for 1 h. The culture medium was then replaced by fresh medium, and the cells were incubated at 37 °C in a 5% CO_2 incubator.

Virus Adsorption/Entry Stage. To examine whether the compound affected viral binding and entry, RD cells were prechilled at 4 °C (which allows the virus to attach to the cell surface but not complete cellular entry) for 1 h, and then the media was replaced by a mixture of AS (0.08 μmol/L) and CVA16 (10^4 $TCID_{50}$/0.1 mL). After incubation at 37 °C for a further 1 h, the cells were washed with PBS, overlaid with 2% MEM, and incubated at 37 °C under 5% CO_2.

Virus Replication Stage. To examine whether the compound affected viral replication, CVA16 (10^4 $TCID_{50}$/0.1 mL) was inoculated with RD cells for 1 h. After the removal of the viral inoculum, media containing 0.08 μmol/L AS was added to the infected cells and incubated at 37 °C for a further 5 h. The media with the compound was discarded, replaced with 2% MEM, and incubated at 37 °C.

Virus Release Stage. RD cells were challenged with CVA16 (10^4 $TCID_{50}$/0.1 mL) in 2% MEM for 1 h, washed with PBS, and then covered by maintenance media. At 5 hpi the culture media was replaced by fresh medium including AS (0.08 μmol/L), and the cells were incubated at 37 °C under 5% CO_2 until 16 hpi. Then the cells were washed with PBS again, covered with fresh medium, and cultured at 37 °C under 5% CO_2 sequentially.

For all of the above experiments examining four stages of the viral life cycle, the CPE was observed and recorded through an inverted microscope every 12 h. Real-time PCR was used to detect the viral RNA in the culture supernatant at 48 h. Total RNA was extracted using Trizol reagent (TaKaRa, Dalian, China) and reverse transcribed to cDNA using a ReverTra Ace qPCR RT kit (Toyobo, Osaka, Japan) on a SimpliAmp thermal cycler (ABI, Singapore), according to the manufacturer's instructions. The cDNA was then subjected to standard end-point PCR amplification employing specific oligonucleotide primers targeting the CVA16 VP1 gene. The forward primer was 5′-TGACCAAACTGTGAACAATCAAGTG-3′, and the reverse primer was 5′-GCATATCCCATCAAATCAATGTCC-3′. The amplicon was gel purified (CWBIO, Xiamen, China) and then cloned to the pMD19-T vector (Takara, Dalian, China) and transformed into *E. coli* DH5α competent cells. The plasmid DNA was quantified using the Nanodrop 2000 UV−vis spectrophotometer (Thermo Fisher, Shanghai, China), copies/μL were calculated, and absolute standards were prepared. An external standard curve was established by real-time PCR with a GoldStarTaqman mixture kit (CWBIO, Beijing, China) on the 7500 fast real-time PCR system (ABI, Singapore). The oligonucleotide primers employed for the qPCR were 5′-ACTGTGAACAATCAAGTGAACC-3′ (forward) and 5′-CGGCTTGTAGTGCTGGTAC-3′ (reverse), and the TaqMan hydrolysis probe was ROX-AGTGCCTAACCTGTGACA-CTTGCCTC-BHQ2 at final concentrations of 0.4 μmol/L (each primer) and 0.4 μmol/L (of probe). The primers and probes employed in the real-time PCR were obtained from the Chinese CDC's "Enterovirus Laboratory Manual". Real-time PCR was carried out under the following thermocycling conditions: 95 °C for 10 min and 40 cycles of 95 °C for 15 s, 60 °C for 60 s with fluorescence signal

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collected and recorded at the end of each cycle. The viral RNA was expressed as copies per 100 μL of the supernatant.

Viral Inactivation Assay. Two 1:1 mixtures of AS (0.08 μmol/L) or 2% MEM and CVA16 (10^4 $TCID_{50}$/0.1 mL) were prepared and denoted as group A and B, respectively. Both groups were preincubated at 37 °C in a 5% CO_2 incubator for 2 h. RD cell monolayers seeded in six-well plates were prechilled at 4 °C for 1 h, and the compound−virus mixture or 2% MEM−virus mixture were separately inoculated to the cells and incubated at 4 °C for another 2 h. Then the cells were washed with precooled PBS and incubated at 37 °C for a further 48 h. A cell control (untreated and uninfected) was also established. Development of CPE was monitored under an inverted microscope at 12, 24, 36, and 48 hpi, and the supernatant at the corresponding time points was collected at 24 and 48 hpi. CVA16 viral loads were measured by real-time PCR, as described above.

Antiviral Test *in Vivo*. *Animals.* Specific-pathogen-free ICR mice were purchased from Jinan Pengyue Experimental Animal Technology Co., Ltd. [Qualification No. 37009200009461], with a total of 6 litters (3 litters: 17 mice/litter; 3 litters: 15 mice/litter).

The animal experiments described here were approved by the Taishan Medical College Administrative Committee for Laboratory Animals (permission no. 2018087) and conformed to the guidance of the Shandong Laboratory Animal Welfare and Ethics of Shandong Administrative Committee of Laboratory Animals. ICR mice feeding corresponded to the national animal welfare and ethical standards.

Compound Treatment Test. Five-day-old ICR neonatal mice were challenged with CVA16 ($10^{5.5}$ $TCID_{50}$/g) by intramuscular injection (i.m.) in hind limb muscles. One hour later, 2 mg/kg AS or sterile cultural media was injected i.m. Daily changes in body weight, behavior, and mortality of the prevention, treatment, negative control, and positive control groups were recorded. The hearts, lungs, contralateral hind limb skeletal muscles, and spinal muscles of the first three groups were removed from the neonates that died or were euthanized at 1, 3, and 5 dpi. Total RNA was extracted from half of the organs using Trizol reagent (TaKaRa, Dalian, China) and subsequently reverse transcribed to cDNA using a ReverTra Ace qPCR RT kit (Toyobo, Osaka, Japan) on a SimpliAmp thermal cycler (ABI, Singapore), in accordance with the manufacturer's instructions. The cDNA was then subjected to real-time PCR to determine the viral loads in the organs and tissues. The other half of the organs were fixed in 10% neutral formalin and paraffin sections and prepared for hematoxylin−eosin (H&E) staining to examine histopathology. At the same time, the cardiac blood of the suckling mice infected with CVA16 was collected at different time points (3 and 5 dpi), and the serum was separated by centrifugation at 3000*g* for 30 min and stored at −20 °C. The expression levels of the pro-inflammatory cytokines IL-6 and IFN-γ were tested by enzyme-linked immunosorbent assay (ELISA) detection kits (Multisciences Biotechnology Co., Ltd., Hangzhou, China). The theoretical limits of detection were as follows: IL-6, 1.17 pg/mL; IFN-γ, 1.74 pg/mL.

Compound Prevention Test. Medium containing 0.08 μmol/L AS was 1:1 mixed with CVA16 solution (10^4 $TCID_{50}$/0.1 mL), and then the mixture was incubated at 37 °C for 2 h. Similarly, the compound−virus mixture was administered i.m. to hind limb muscles of 5-day-old mice exactly as was performed for the compound treatment test. The body weights, clinical symptoms, and survival rates of the mice were monitored daily. In addition, the hearts, lungs, hind limbs, and spinal muscles of suckling mice in different groups were collected to analyze the virus load by real-time PCR, and histopathological analysis was performed as described earlier. When the mice in the positive group were near death, the serum of the suckling mice was collected to measure the expression levels of IFN-γ and IL-6 according to the instructions of the ELISA kit.

Statistical Analysis. Statistical analysis was performed using GraphPad Prism 7 (GraphPad 7 Software, San Diego, CA, USA). Mantel−Cox log rank software was used to evaluate the survival rate and mortality of the suckling mice in each group. Viral loads were measured using the two-tailed Mann−Whitney U test. The data were expressed as mean ± standard deviation. The means between the groups were compared using analysis of variance and processed with SPSS version 20.0 software. A P-value of <0.05 after two-tailed t testing was regarded as significantly different. $P < 0.05$ is denoted by *, $P < 0.01$ by **, and $P < 0.001$ by ***. $P > 0.05$ indicates no significant difference, denoted by N.S.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jnatprod.8b00735.

Figure S1 (PDF)
Figure S2 (PDF)
Figure S3 (PDF)

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Notes

The authors declare no competing financial interest.

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